

# Recovering and Redirecting Peptide Backbone Conformation via Contact-Conditioned C $\alpha$ Generation in a Cross-Reactive pMHC/TCR System

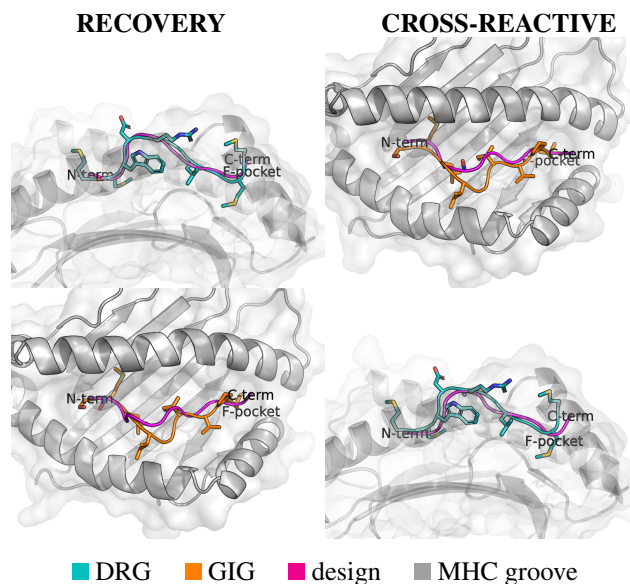
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**Motivation.** Some T-cell receptors are cross-reactive: they recognize several different peptides by letting each peptide fold into a different shape inside the same MHC groove [2]. This cross-reactivity is a central safety problem for TCR-based therapies, where an engineered receptor can cause dangerous off-target effects by recognizing an unintended self-peptide that adopts a compatible backbone shape. Deep generative models can now build protein backbones with high accuracy (RFDiffusion [1]), which raises a question that bears directly on that problem: given only the receptor side of the interaction, can such a model be steered to build a peptide backbone in one chosen target shape, even a shape that is not the one closest to its input? We test this in a well-studied two-peptide system [2]: the DMF5 receptor on HLA-A\*02:01 binds two 10-residue peptides, GIG (6AM5) and DRG (6AMU), whose backbones differ by 2.87 Å almost entirely at the tail end (the first four positions match within 1.14 Å; position 6 moves 5.62 Å). Which register a peptide takes is set by which residue drops into the MHC F-pocket (position 10 for GIG, position 9 for DRG).

**Methods.** Conditioning only on receptor-side contact residues, with no peptide-backbone template, we asked whether RFDiffusion [1] can generate a C $\alpha$  backbone that recovers a peptide’s own native register or crosses into the alternate one. We generated 300 designs per crystal for 20 different conditioning schemes, plus a focused N-terminus-seeding experiment. Designs are scored against a three-criterion register test calibrated not to a single crystal but to **native molecular-dynamics ensembles** (Amber ff19SB, TIP3P, physiological **310 K**, 50 ns): C $\alpha$  proximity within a moving-block-bootstrap acceptance band (DRG within 1.48 Å, 95% CI 1.36 to 1.58), correct F-pocket occupant identity, and native burial depth. We additionally correct for a forward/reverse backbone-threading artifact that raw RMSD cannot detect.

**Results.** Both recovery and redirection are achievable but rare, and are gated by the breadth of conditioning rather than by any specific curated residue set. Across the de novo corpus, 4 of 8,722 designs recover the native DRG register (best 1.07 Å) and 3 of 8,722 cross into the alternate DRG register (best 1.44 Å), all forward-threaded, seating the correct p9 anchor, and from the broadest schemes. Critically, our no-conditioning control produced **zero** designs that dock in the groove or come close in RMSD to either native backbone: **0 of 932** unconditioned draws entered the band or seated an anchor, so these events are real signal above a zero-

information background, not noise. A templating control that supplies backbone geometry directly recovers register on demand (best 0.10 Å), whereas contact conditioning alone only weakly determines it, locating register in a confined tail-end signal. These results, from a single well-characterized cross-reactive system, map the current capability boundary of contact-conditioned backbone generation, separate what it can retrieve from what must still be supplied as explicit geometry, and mark a step toward anticipating the cross-reactive, off-target peptide recognition that constrains the safety of TCR-based therapies.



**Fig 1.** Design (magenta) vs both registers. Left (recovery): matches DRG (top), misses GIG (bottom). Right (cross-reactive): departs GIG (top), crosses into DRG (bottom).

**Table 1:** De novo and control designs, scored against the 310 K native envelope. Recoveries land in the peptide’s own native register; crossings land in the alternate register.

| design group                | <i>n</i> | best RMSD<br>own (Å) | recov.             | cross. |
|-----------------------------|----------|----------------------|--------------------|--------|
| de novo (receptor contacts) | 8,722    | 1.07                 | 4                  | 3      |
| null (no conditioning)      | 932      | 2.05                 | 0                  | 0      |
| templated (backbone given)  | 3,060    | 0.10                 | register on demand |        |
| crossover (N-term given)    | 338      | 0.54                 | 16                 | 20     |

**References.** [1] Watson JL et al. De novo design of protein structure and function with RFDiffusion. *Nature* 620:1089–1100 (2023). [2] Riley TP et al. T cell receptor cross-reactivity expanded by dramatic peptide-MHC adaptability. *Nat. Chem. Biol.* 14:934–942 (2018).

**Availability.** Code, MD-calibrated scorer, and curated design archives: [github.com/sermare/if-mhc](https://github.com/sermare/if-mhc).